

Custom wake-up response

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1. Course Content

- Adding audio files to the multi_brains program audio library allows you to customize the wake-up voice response.

[!NOTE]

- This tutorial is only for users who need to customize personalized voice responses and does not affect normal use.
- If you do not need to customize personalized responses, you can skip this section.

2. Preparing Audio Files

Note: You need to copy the `generate_voice` file from the "Source Code" folder to the current board for testing. For Jetson Nano and Raspberry Pi 5 boards, you will need to use Docker.

- First, place the `generate_vioce` file from the source code in the root directory. Then, place the command in the following path:

```
sudo cp generate_vioce /usr/local/bin/
```

- You can prepare the audio material for the voice reply yourself.
- Alternatively, you can generate the voice using the system's built-in `generate_vioce` command. The voice generation model uses the Alibaba Cloud Model Studio Platform's speech synthesis model, so you need to configure `ALIYUN_API_KEY` according to the tutorial in [02-Configure API-KEY](#) of this chapter first.
- Pre-configure Alibaba Cloud Model Studio Platform API-KEY

```

! multi_brains_setting.yaml X
D: > ! multi_brains_setting.yaml
1 #####
2 #common setting
3 #国内外通用设置项 Common settings applicable both domestically and internationally
4 #####
5 LANGUAGE: 'en' #用户选择: zh:中国用户 en:International users
6 DEBUG_MODE: False #调试模式
7
8 #####
9 # dify setting
10 # dify配置选项
11 #####
12 DIFY_BASE_URL: "http://localhost/v1"
13
14 #####中国用户#####
15 # nuwa-camera
16 #DIFY_API_KEY: "app-Jbp4IptHkD9kwZjXwYU0X0IYP" #dify多智能体大模型应用API_KEY
17 # usb-camera
18 #DIFY_API_KEY: "app-hgY2jMhFJKTbpuC4ZwiJVhm" #dify多智能体大模型应用API_KEY
19
20 ##### International users #####
21 # nuwa-camera
22 #DIFY_API_KEY: "app-5IZ73phJF7ILyPu4Lw3dR0In" #dify Multi-Agent Large Model Application API_KEY
23 # usb-camera
24 #DIFY_API_KEY: "app-uB1RKhDCy2TLdbQ6zFSUuVY" #dify Multi-Agent Large Model Application API_KEY
25
26
27
28 #####
29 #ASR function setting
30 #语音识别功能设置
31 #####
32 ALIYUN_API_KEY: 'sk-*****' #阿里云API_KEY International users do not need to fill in this
33 USE_ONLINE_ASR : True #是否使用在线 ASR Use online ASR?
34 ASR_SUPPLIER : 'xunfei' #ASR供应商 (仅当使用在线ASR时生效): 中国大陆地区: aliyun 国际地区: xunfei #ASR Provider (Only a
35 ONLINE_ASR_MODEL : 'paraformer-realtime-v2'
36 ASR_THRESHOLD : 3 #ASR识别结果阈值, 低于该阈值则不进行后续处理, 单位: 字符 ASR recognition result threshold; if t
37 WAKEUP_THRESHOLD : 5.0 #唤醒时间阈值, 防止WAKEUP_THRESHOLD时间内被多次唤醒, 单位: 秒 Wake-up time threshold to preve
38 VAD_MODE: 2 #vad灵敏度, 使用频率较少, 可以先调整麦克风灵敏度硬件灵敏度# VAD sensitivity
39 MAX_SILENCE_FRAMES: 90 #尾音时长检测, 单位: 帧数, 1帧~30毫秒 End-of-sound duration detection, unit: frames, 1 frame
40

```

- Run the command in the terminal:

```
generate_vioce --text Hello --language_type English
```

- Where `--text` is the startup parameter, "Hello" Replace with the content you want to synthesize the speech for.

```

jetson@yahboom:~/Downloads/dist$ generate_vioce --text 你好
音频文件已保存至: /home/jetson/audio.wav
jetson@yahboom:~/Downloads/dist$

```

- Audio files will be automatically saved in the `~/generate_voices/` directory.

[!NOTE]

Other optional startup parameters are as follows:

- `--voice`: Speaker, default `Cherry`
- `--language_type`: Language, specify the language of the synthesized audio. Available options include `Chinese`, `English`, `German`, `Italian`, `Portuguese`, `Spanish`, `Japanese`, `Korean`, `French`, and `Russian`.
- `--save_path`: Audio file save path, default `~/`
- `--config_file`: Configuration file path, default `~/yahboomcar_ros2_ws/yahboomcar_ws/src/multi_brains/config/multi_brains_setting.yaml`
- `--text`: Text to be synthesized audio, default is empty
- `--model` The speech synthesis model defaults to `qwen3-tts-flash`.

For available speakers and speech synthesis models, please refer to the dynamic notifications on the Bailian official website: https://bailian.console.aliyun.com/?spm=5176.29619931.J_SEsJsNv72y.RuRFS2VknO.2.74cd10d73l2Pw5&tab=doc#/doc?type=model&url=2879134

- Reference Model:

Model availability

We recommend Qwen3-TTS-Flash.

Qwen3-TTS-Flash offers 49 voices and supports multiple languages and dialects.

Qwen-TTS offers up to 7 voices and supports only Chinese and English.

International (Singapore)China (Beijing)	
Model	Version
qwen3-tts-flash Capabilities are identical to qwen3-tts-flash-2025-09-18.	Stable
qwen3-tts-flash-2025-11-27	Snapshot
qwen3-tts-flash-2025-09-18	Snapshot

- Reference Voice

Qwen3-TTS-Flash

Supported voices vary by models. Set the `voice` request parameter

View supported voices of qwen3-tts-flash-2025-11-27

Name	<code>voice</code> parameter
Cherry	Cherry
Serena	Serena
Serena	Ethan
Chelsie	Chelsie
Momo	Momo
Vivian	Vivian
Moon	Moon

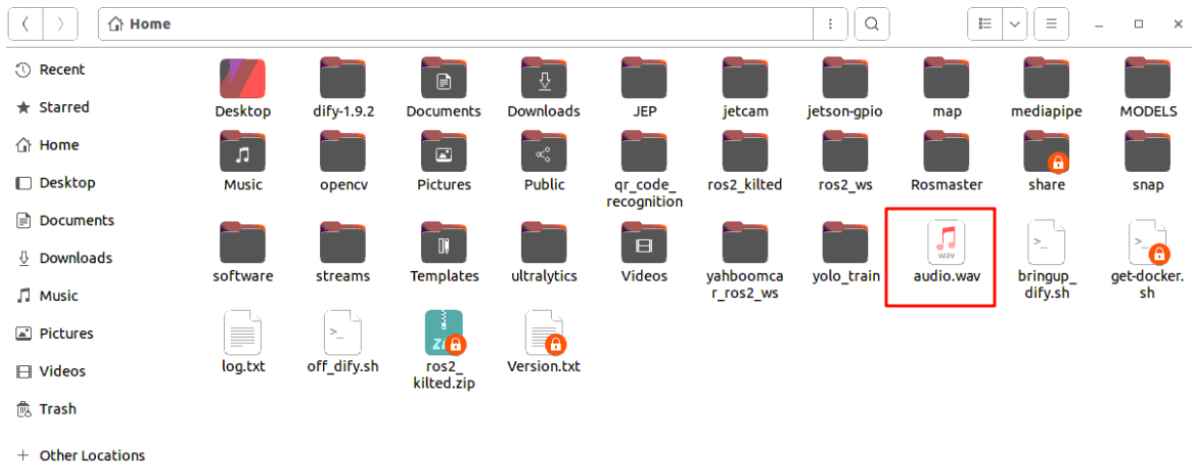
- Supported Languages

Chinese, English, Spanish, Russian, Italian, French, Korean, Japanese, German, Portuguese

3. Loading Audio Files

- Multi_brains system audio path:

```
/home/jetson/
```



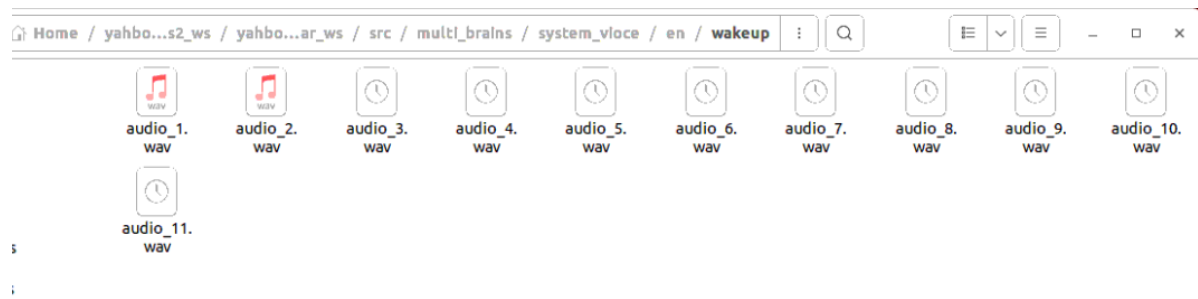
- Where:
- **zh** is the Chinese response audio, suitable for Chinese users. Place the prepared audio files in this directory:

```
~/yahboomcar_ros2_ws/yahboomcar_ws/src/multi_brains/system_vioce/zh/wakeup
```



- **en** is the English response audio, suitable for international users. Place the prepared audio files in this directory:

```
~/yahboomcar_ros2_ws/yahboomcar_ws/src/multi_brains/system_vioce/en/wakeup
```



tions

- When the multi_brains program is launched, the audio file in the corresponding directory will be automatically loaded, and a personalized response voice will be randomly played when the user wakes up.